EXPERIMENT – 7

HANDLING OUTLIERS & MISSING VALUES IN A DATASET

AIM:

To preprocess the dataset by removing duplicates, detecting outliers, and handling missing values using Pandas and Scikit-learn SimpleImputer

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. A dataset for Practice
3. Jupyter Notebook
4. Knowledge on Pandas & Scikit-Learn Libraries

Step 1: Import the basic libraries

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load & Analyze the Dataset

In [ ]:
```
# Load the student dataset
df_241UDS0092 = pd.read_excel("/content/research_student (1).xlsx")

# Dataset Information
df_241UDS0092.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  220 non-null    object
 1   Marks[10th]             220 non-null    float64
 2   Marks[12th]             220 non-null    float64
 3   Gender                  220 non-null    object
 4   Board[10th]             220 non-null    object
 5   Board[12th]             220 non-null    object
 6   Category                220 non-null    object
 7   GPA 1                   220 non-null    float64
 8   Rank                    221 non-null    float64
 9   Normalized Rank         220 non-null    float64
 10  CGPA                    220 non-null    float64
 11  Current Back            220 non-null    float64
 12  Ever Back               219 non-null    float64
 13  GPA 2                   220 non-null    float64
 14  GPA 3                   220 non-null    float64
 15  GPA 4                   220 non-null    float64
 16  GPA 5                   220 non-null    float64
 17  GPA 6                   220 non-null    float64
 18  Olympiads Qualified     220 non-null    float64
 19  Technical Projects      220 non-null    float64
 20  Tech Quiz               220 non-null    float64
 21  Engg. Coaching          220 non-null    float64
 22  NTSE Scholarships       220 non-null    float64
 23  Miscellany Tech Events  220 non-null    float64
dtypes: float64(19), object(5)
memory usage: 41.9+ KB
```

In [ ]:
```
#Display the first five records
df_241UDS0092.head()
```

Out[8]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	CIVIL	77.57	64.6	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	86.40	71.8	Male	CBSE	CBSE	GEN
3	CSE	88.14	78.0	Male	ICSE	ICSE	GEN
4	CSE	65.40	59.8	Female	CBSE	CBSE	ST

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	NaN	NaN	NaN	...	NaN	NaN	NaN
1	6.29	44718.0	15.970714	...	5.94	5.41	6.25
2	6.47	24222.0	8.650714	...	5.88	5.53	6.44
3	7.35	24723.0	8.829643	...	6.54	6.41	6.50
4	6.41	232157.0	82.913214	...	5.71	5.24	5.88

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0
4	6.25	2.0	0.0	2.0	2.0	0.0	1.0

5 rows × 24 columns

In [ ]:
```
#Check the shape
df_241UDS0092.shape
```

Out[9]: (223, 24)

241UDS0092

In [ ]:
```
# Remove completely empty rows
df_241UDS0092 = df_241UDS0092.dropna(how='all')

df_241UDS0092.shape
```

Out[10]: (221, 24)

Step 3: Find & Remove the Duplicates

In [ ]:
```
# Checking the number of duplicates
df_241UDS0092[
    ['Marks[10th]', 'Marks[12th]', 'CGPA']
].duplicated().sum()
```

Out[12]: np.int64(0)

In [ ]:
```
# Remove the duplicates
df_241UDS0092 = df_241UDS0092.drop_duplicates()

df_241UDS0092.shape
```

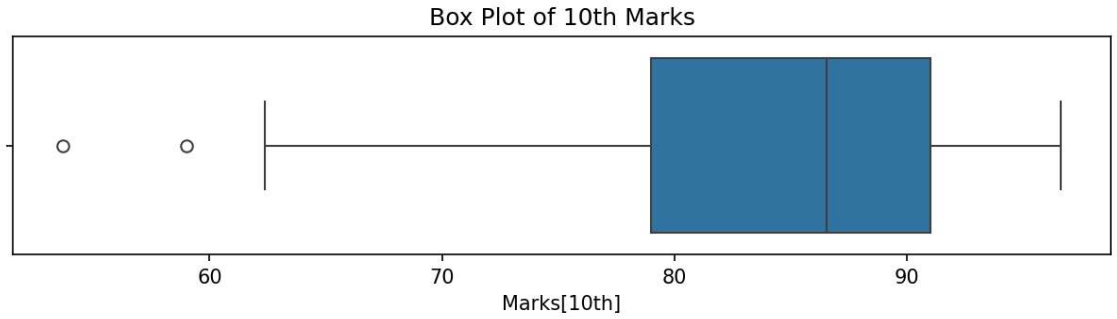
Out[13]: (221, 24)

Step 4: Identify & Remove the Outliers

In [ ]:
```
plt.figure(figsize=(10, 2), dpi=150)

sns.boxplot(
    data=df_241UDS0092['Marks[10th]'],
    orient='h'
)

plt.title('Box Plot of 10th Marks')
plt.show()
```

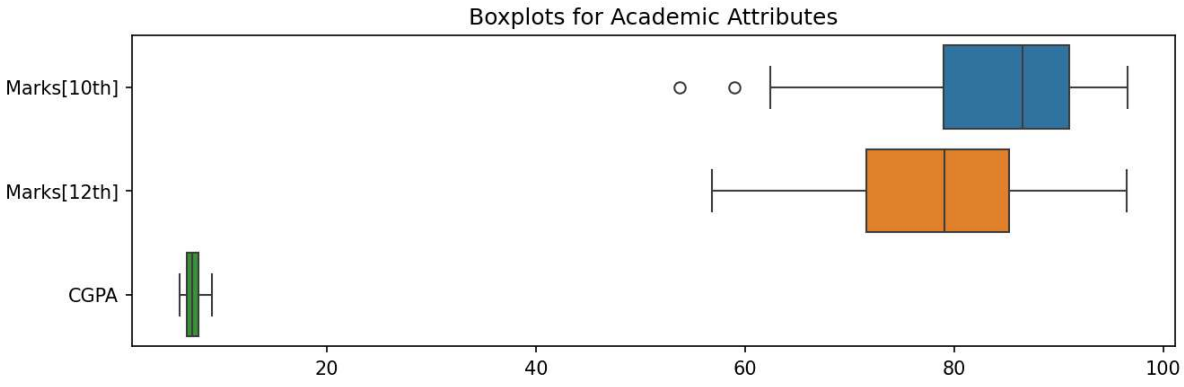


In [ ]:
```
#Create a list for outlier columns
columns_241UDS0092 = [
    'Marks[10th]',
    'Marks[12th]',
    'CGPA'
]
```

In [ ]:
```
#Visualization with Boxplots
plt.figure(figsize=(10, 3), dpi=150)

sns.boxplot(
    data=df_241UDS0092[columns_241UDS0092],
    orient='h'
)

plt.title('Boxplots for Academic Attributes')
plt.show()
```



In [ ]:
```
#Calculate the IQR Ranges Q1 and Q3
Q1_241UDS0092 = df_241UDS0092[
    columns_241UDS0092
].quantile(0.25)

Q3_241UDS0092 = df_241UDS0092[
    columns_241UDS0092
].quantile(0.75)

Q1_241UDS0092, Q3_241UDS0092
```

Out[18]:
```
(Marks[10th]    79.0000
 Marks[12th]    71.5500
 CGPA            6.6375
 Name: 0.25, dtype: float64,
 Marks[10th]    91.0000
 Marks[12th]    85.2500
 CGPA            7.7025
 Name: 0.75, dtype: float64)
```

In [ ]:
```
#Calculate IQR
IQR_241UDS0092 = Q3_241UDS0092 - Q1_241UDS0092

IQR_241UDS0092
```

Out[19]:

	0
Marks[10th]	12.000
Marks[12th]	13.700
CGPA	1.065

dtype: float64

In [ ]:
```
#Find the Outliers using IQR Range
outliers_241UDS0092 = (
    (df_241UDS0092[columns_241UDS0092] <
     (Q1_241UDS0100 - 1.5 * IQR_241UDS0100))
    |
    (df_241UDS0092[columns_241UDS0092] >
     (Q3_241UDS0092 + 1.5 * IQR_241UDS0092))
)
```

In [ ]:
```
outliers_241UDS0092.sum()
```

Out[21]:

	0
Marks[10th]	2
Marks[12th]	0
CGPA	0

dtype: int64

In [ ]:
```
df_241UDS0092.shape
```

Out[22]:
```
(221, 24)
```

In [ ]:
```
# Removing the outliers
df_241UDS0092 = df_241UDS0092[
    ~(outliers_241UDS0092).any(axis=1)
]
```

In [ ]:
```
df_241UDS0092.head()
```

Out[24]:

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
1	CIVIL	77.57	64.6	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	86.40	71.8	Male	CBSE	CBSE	GEN
3	CSE	88.14	78.0	Male	ICSE	ICSE	GEN
4	CSE	65.40	59.8	Female	CBSE	CBSE	ST
5	CSE	81.00	74.0	Male	CBSE	CBSE	GEN

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
1	6.29	44718.0	15.970714	...	5.94	5.41	6.25
2	6.47	24222.0	8.650714	...	5.88	5.53	6.44
3	7.35	24723.0	8.829643	...	6.54	6.41	6.50
4	6.41	232157.0	82.913214	...	5.71	5.24	5.88
5	6.80	23252.0	8.304286	...	5.88	6.00	5.93

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0
4	6.25	2.0	0.0	2.0	2.0	0.0	1.0
5	5.44	2.0	0.0	2.0	2.0	0.0	1.0

5 rows × 24 columns

Step 5: Fill the Missing Values

In [ ]:
```
# Find any missing values
df_241UDS0092.isnull().sum()
```

Out[25]:

	0
Branch	1
Marks[10th]	1
Marks[12th]	1
Gender	1
Board[10th]	1
Board[12th]	1
Category	1
GPA 1	1
Rank	0
Normalized Rank	1
CGPA	1
Current Back	1
Ever Back	2
GPA 2	1
GPA 3	1
GPA 4	1
GPA 5	1
GPA 6	1
Olympiads Qualified	1
Technical Projects	1
Tech Quiz	1
Engg. Coaching	1
NTSE Scholarships	1
Miscellany Tech Events	1

dtype: int64

In [ ]:
```
# Filling Marks[10th] with mean value
df_241UDS0092['Marks[10th]'] = df_241UDS0092[
    'Marks[10th]'
].fillna(
    np.mean(df_241UDS0092['Marks[10th]'])
)
```

In [ ]:
```
df_241UDS0092['Marks[10th]'].isnull().sum()
```

Out[27]:
```
np.int64(0)
```

In [ ]:
```
df_241UDS0092['Marks[10th]'].isnull().sum()
```

Out[28]:
```
np.int64(0)
```

In [ ]:
```
#Import SimpleImputer
from sklearn.impute import SimpleImputer
```

In [ ]:
```
#Create the Imputer
numerical_imputer_241UDS0100 = SimpleImputer(
    strategy='median'
)
```

In [ ]:
```
#Apply Median Imputation
df_241UDS0092[
    ['Marks[12th]', 'CGPA']
] = numerical_imputer_241UDS0092.fit_transform(
    df_241UDS0092[['Marks[12th]', 'CGPA']]
)
```

In [ ]:
```
#Check Missing Values
df_241UDS0092[
    ['Marks[12th]', 'CGPA']
].isnull().sum()
```

Out[32]:

	0
Marks[12th]	0
CGPA	0

dtype: int64

Filling Branch with Mode using SimpleImputer

In [ ]:
```
#First check the branch frequencies:

df_241UDS0092['Branch'].value_counts()
```

Out[33]:

Branch	count
ECE	52
EEE	50
CSE	46
MECH	27
CIVIL	18
IT	16
PROD	9

dtype: int64

Create Categorical Imputer

In [ ]:
```
categorical_imputer_241UDS0092 = SimpleImputer(
    strategy='most_frequent'
)
```

Impute Missing Branch Values

In [ ]:
```
df_241UDS0092[['Branch']] = categorical_imputer_241UDS0092.fit_transform(
    df_241UDS0092[['Branch']]
)
```

Check the Branch Values

In [ ]:
```
df_241UDS0092['Branch'].value_counts()
```

Out[36]:

Branch	count
ECE	53
EEE	50
CSE	46
MECH	27
CIVIL	18
IT	16
PROD	9

dtype: int64

In [ ]:
```
# Cleaned Dataset Information
df_241UDS0092.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 219 entries, 1 to 222
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  219 non-null    object
 1   Marks[10th]             219 non-null    float64
 2   Marks[12th]             219 non-null    float64
 3   Gender                  218 non-null    object
 4   Board[10th]             218 non-null    object
 5   Board[12th]             218 non-null    object
 6   Category                218 non-null    object
 7   GPA 1                   218 non-null    float64
 8   Rank                    219 non-null    float64
 9   Normalized Rank         218 non-null    float64
 10  CGPA                    219 non-null    float64
 11  Current Back            218 non-null    float64
 12  Ever Back               217 non-null    float64
 13  GPA 2                   218 non-null    float64
 14  GPA 3                   218 non-null    float64
 15  GPA 4                   218 non-null    float64
 16  GPA 5                   218 non-null    float64
 17  GPA 6                   218 non-null    float64
 18  Olympiads Qualified     218 non-null    float64
 19  Technical Projects      218 non-null    float64
 20  Tech Quiz               218 non-null    float64
 21  Engg. Coaching          218 non-null    float64
 22  NTSE Scholarships       218 non-null    float64
 23  Miscellany Tech Events  218 non-null    float64
dtypes: float64(19), object(5)
memory usage: 42.8+ KB
```

Check all remaining missing values

In [ ]:
```
df_241UDS0092.isnull().sum()
```

Out[38]:

	0
Branch	0
Marks[10th]	0
Marks[12th]	0
Gender	1
Board[10th]	1
Board[12th]	1
Category	1
GPA 1	1
Rank	0
Normalized Rank	1
CGPA	0
Current Back	1
Ever Back	2
GPA 2	1
GPA 3	1
GPA 4	1
GPA 5	1
GPA 6	1
Olympiads Qualified	1
Technical Projects	1
Tech Quiz	1
Engg. Coaching	1
NTSE Scholarships	1
Miscellany Tech Events	1

dtype: int64

RESULT:

By using Pandas and Scikit-learn, duplicates were removed, outliers were detected and dropped using the IQR method, and missing values were filled using mean, median, and most frequent methods. The dataset is now clean and ready for further analysis.